

Sequences and Series



Sequence limits

1 Determine whether each sequence converges, and find the limit if it does:

a $a_n = \frac{3n+1}{n+2}$

b $a_n = \frac{n^2}{2^n}$

c $a_n = (-1)^n$

2 Evaluate $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n$.

Series and partial sums

3 Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.

4 Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

5 Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.